GlucoTwin

A digital twin for type-2 diabetes that sees glucose spikes coming up to 2 hours ahead, explains why, and lets doctors test what would prevent them

Team Gym Bro · IIIT Delhi

Digital Twin Challenge 2026 · Happiest Health Summit, Bengaluru

2 h

early warning of spikes > 180 mg/dL

300

patients: 200 synthetic Indian + 100 real (ShanghaiT2DM)

0.96

AUROC on held-out synthetic patients

India's diabetes burden is daily, but care is quarterly

101 M

adults in India live with diabetes (ICMR-INDIAB, 2023)

136 M

more have prediabetes - the pipeline keeps growing

~4 / yr

HbA1c checks: a 3-month average that hides daily spikes

288 / day

readings from a CGM sensor - signal that is rarely acted on

Why spikes matter

- Post-meal glucose spikes are common in Indian diets (high-GI rice, roti, chai with sugar, late dinners)
- Repeated hyperglycaemia is linked to vascular complications, yet clinicians see it only weeks later
- Doctors cannot watch hundreds of CGM traces - they need early, explained, prioritised alerts

Our challenge question

Can a virtual replica of each patient - built from their EHR and continuously updated from wearables - predict a new glucose spike above 180 mg/dL up to 2 hours ahead, early and accurately enough for a care team to act?

The use case: predict, explain, act

Predict

Every 5 minutes the twin estimates the probability of a new spike above 180 mg/dL within 2 hours, plus 30/60/120-min glucose forecasts with 80% intervals.

Explain

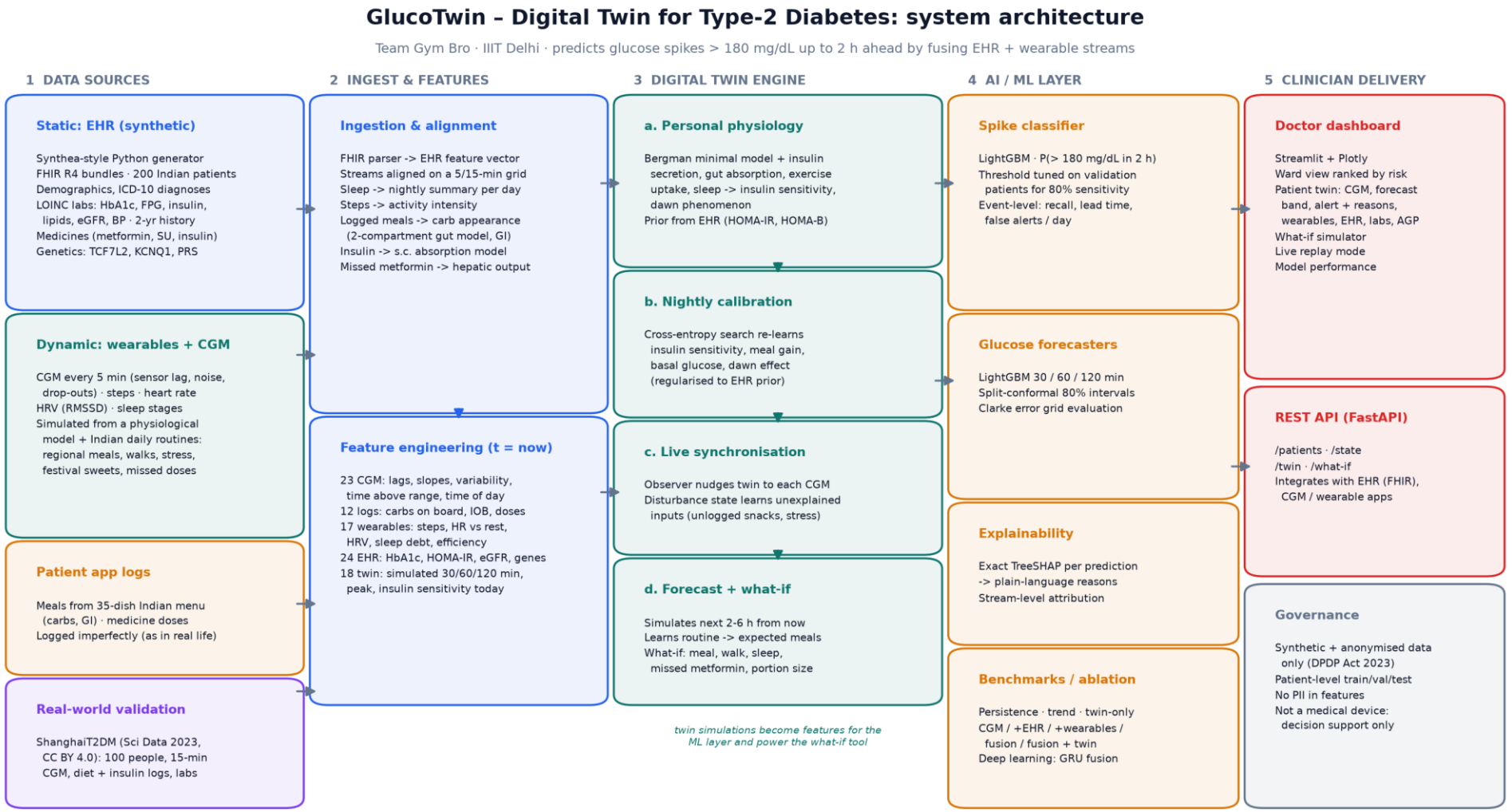
Each alert lists its drivers in plain language - e.g. a rice-heavy lunch, 4.5 h of sleep, no walk, high insulin resistance - from exact SHAP contributions.

Act

The doctor or diabetes educator tests options on the virtual patient - smaller portion, millet swap, 15-min walk - and counsels with evidence.

Who uses it: endocrinologists, physicians and diabetes educators running remote-monitoring programmes for patients who wear a CGM and a smartwatch. Decision support only - not automated dosing.

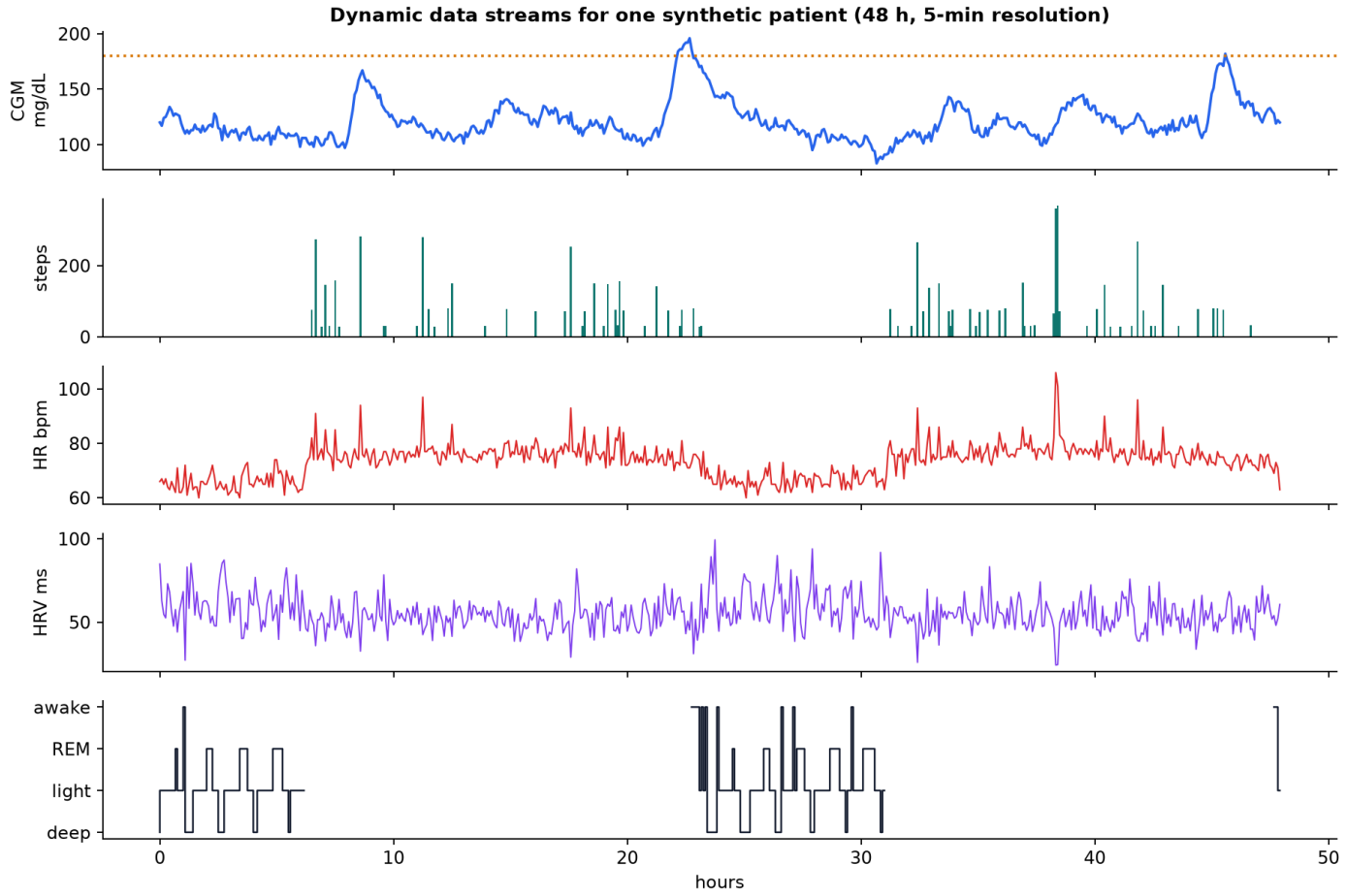
Architecture: five layers, from raw data to clinician



Two data streams, fused on one timeline

Static: EHR (synthetic)

- 200 virtual Indian patients, FHIR R4 bundles
- ICD-10 diagnoses, LOINC-coded labs + 2-year history
- Medicines: metformin, sulfonylurea, DPP-4, premixed insulin
- Genetic markers: TCF7L2, KCNQ1, polygenic risk
- Real-world track: ShanghaiT2DM clinical summaries (100 people)



Dynamic stream (5-min): CGM · steps · heart rate · HRV · sleep stages + meal & medicine logs

Built for Indian patients, not adapted from elsewhere

Regional menus

35 dishes - idli, poha, paratha, rajma chawal, ragi mudde, biryani - with carbs and glycaemic index

Asian phenotype

Asian BMI cut-offs and the 'thin-fat' profile: high insulin resistance at modest BMI

Indian prescribing

Metformin, glimepiride, teneligliptin, premixed 30/70 insulin, as commonly prescribed

Daily rhythm

Sweet chai at 5 pm, dinner after 8.30 pm, weekend shifts, desk-job stress days

Festivals & adherence

Mithai days, missed metformin doses and imperfect meal logging

Genetics

TCF7L2 and KCNQ1 variants relevant in South Asians lower beta-cell function

The twin: a personal physiology that keeps learning

1 Prior from the EHR

Fasting glucose and insulin (HOMA-IR, HOMA-B), weight, HbA1c, medicines and genotype initialise a glucose-insulin model

2 Nightly calibration

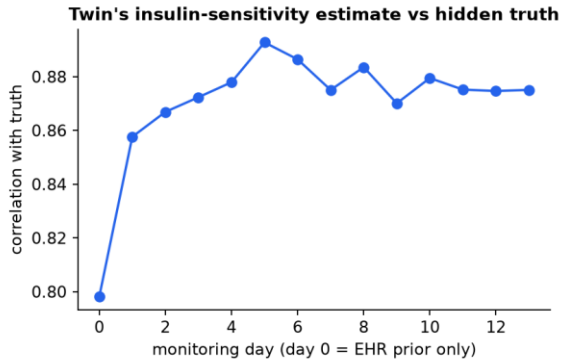
A cross-entropy search re-learns insulin sensitivity, meal response, basal glucose and dawn effect from yesterday's CGM

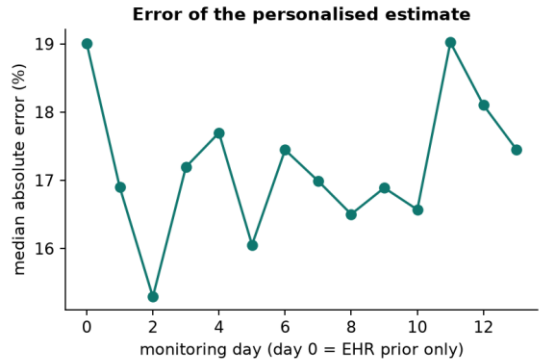
3 Live synchronisation

An observer nudges the twin to every CGM reading and learns unexplained inputs such as unlogged snacks or stress

4 Forecast & what-if

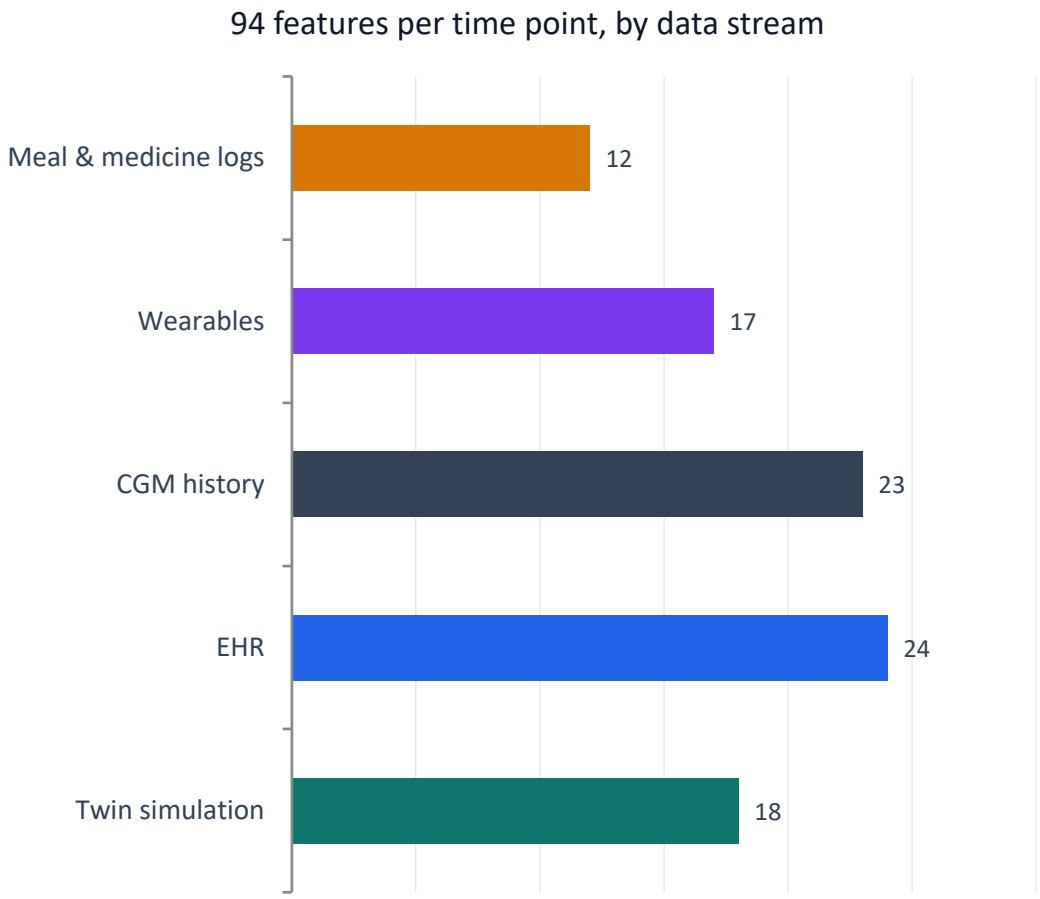
From the synced state the twin simulates the next hours using its learned routine - or any change a doctor proposes





Does it learn the hidden patient?
Each virtual patient has a hidden true insulin sensitivity. The EHR alone predicts it with correlation 0.80; with nightly calibration the twin averages 0.88 and its median error falls from 19.0% to 17.1% (days 1-13). On real patients the synchronised twin alone already beats a "no change" forecast at every horizon.

AI layer: fusion, honest uncertainty, explanations



LightGBM spike classifier
P(new spike > 180 mg/dL within 2 h); threshold tuned on validation patients for 80% sensitivity; 60-min snooze against alarm fatigue

LightGBM glucose forecasters
30 / 60 / 120 min ahead with split-conformal 80% prediction intervals

Exact TreeSHAP explanations
per-prediction contributions grouped and translated into clinician language

Fair comparisons
patient-level splits; baselines (persistence, trend, twin-only), an ablation per data stream and a deep-learning GRU fusion model

Results: spikes flagged early on unseen patients

0.957

AUROC - new spike > 180 mg/dL within 2 h

110 min

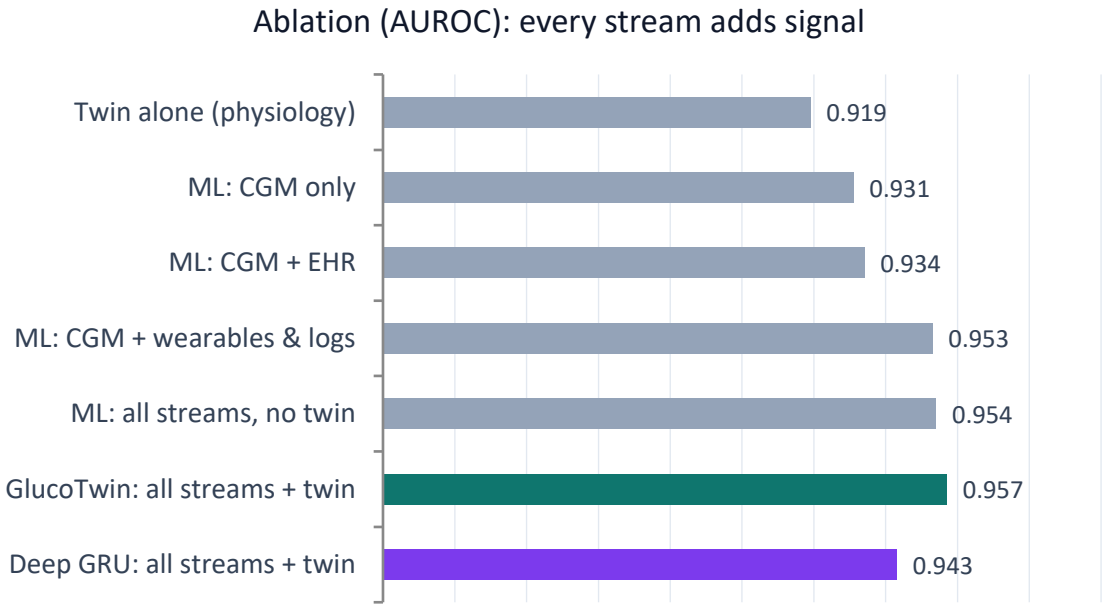
median warning before glucose crosses 180

99%

of 804 spike events flagged >= 15 min ahead

2.7 / day

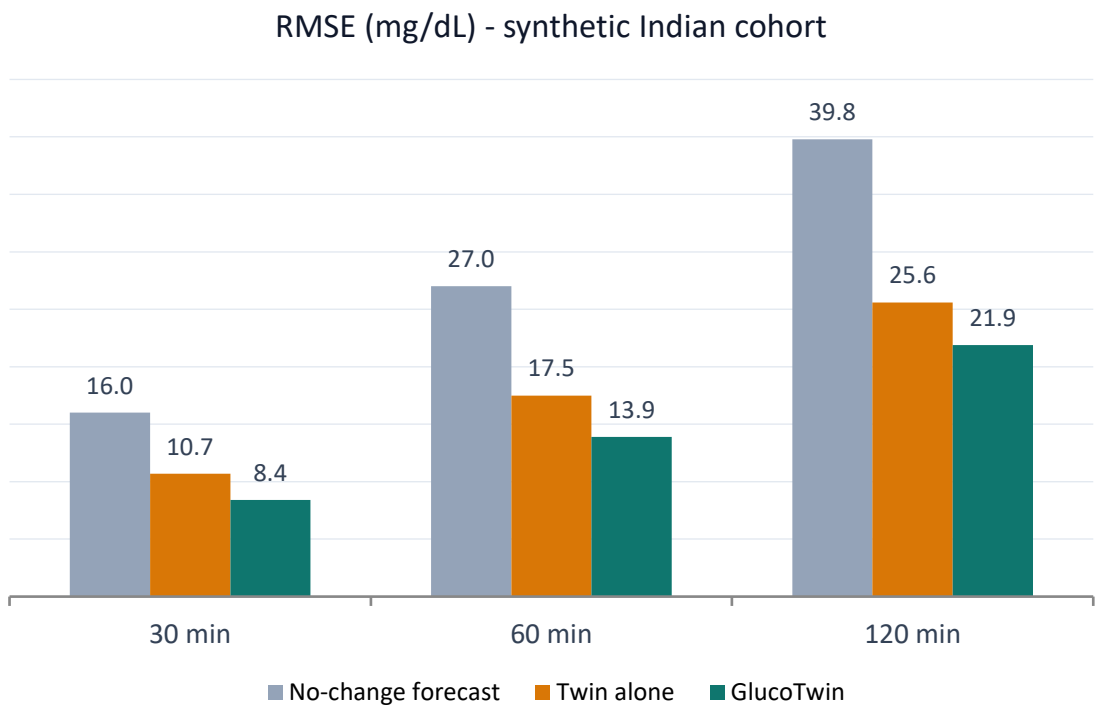
false alerts per patient-day (2.2 real spikes / day)

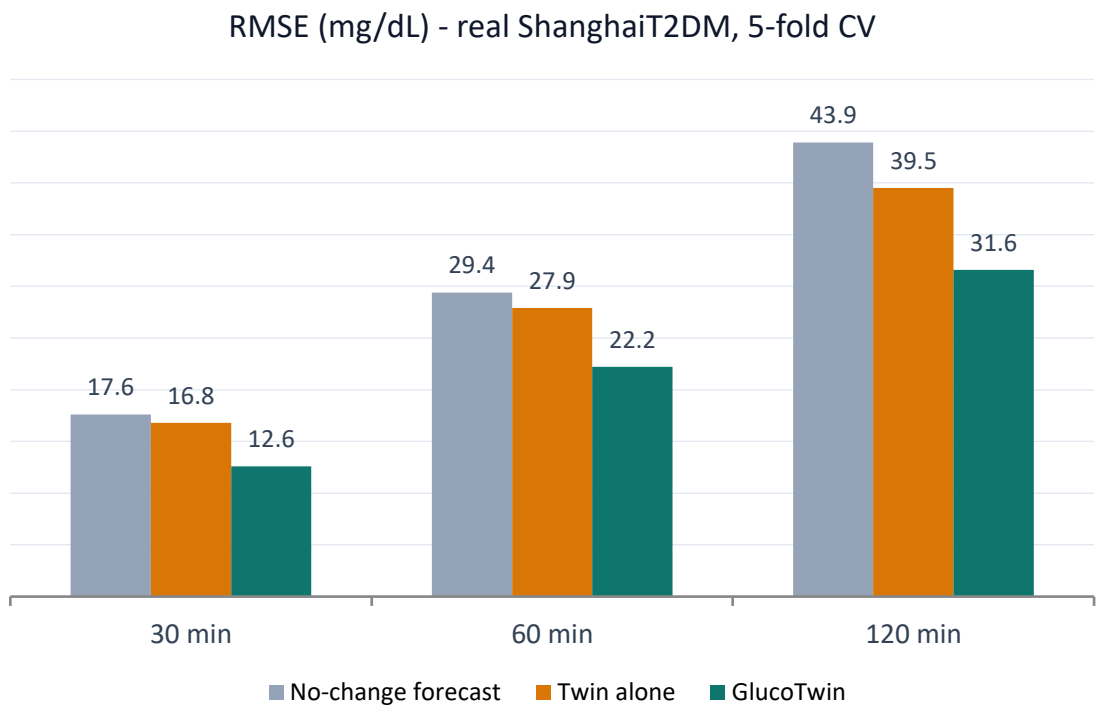


Synthetic Indian cohort: 30 held-out patients, 103,680 time points. Trend-rule baseline AUROC 0.706. Sensitivity 84% at specificity 91%.

Takeaway: CGM alone is good, but adding wearables, logs and the twin's simulation lifts discrimination and cuts false alarms. The twin's simulated peak is the single most important feature.

Forecasts: about half the error of a naive guess





-48%

1-h forecast error vs no-change (synthetic)

96.7%

Clarke zone A at 60 min (synthetic)

-25%

1-h forecast error vs no-change (real patients)

81%

real coverage of the 80% interval (60 min)

Real-world check: 100 people with T2D

0.878

AUROC, 5-fold CV grouped by person

22.2

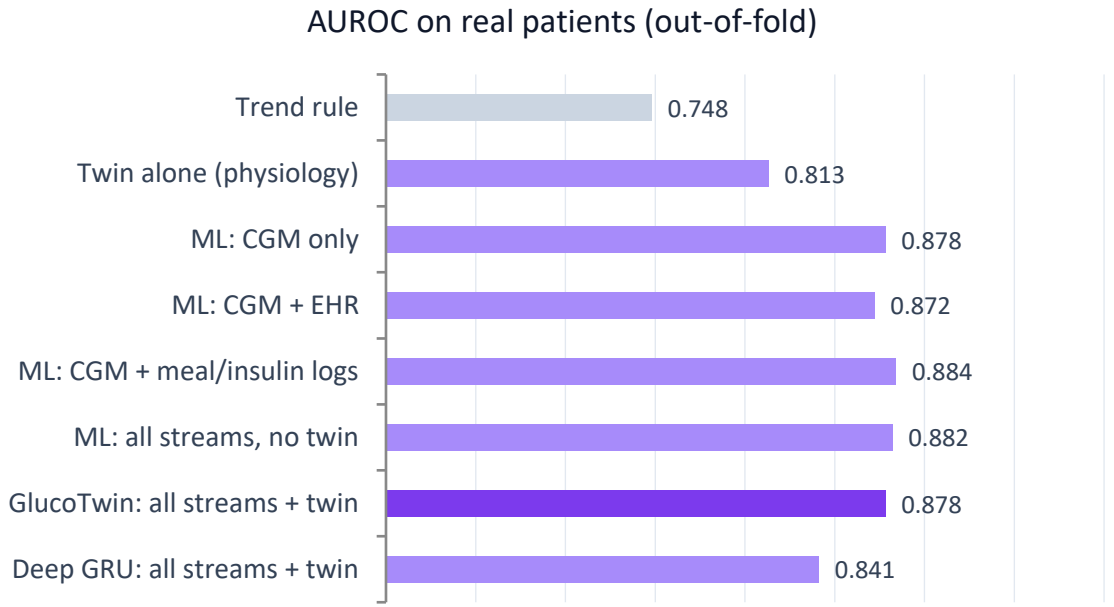
mg/dL 1-h RMSE (no-change: 29.4)

99%

of 1823 real spike events flagged >= 15 min ahead

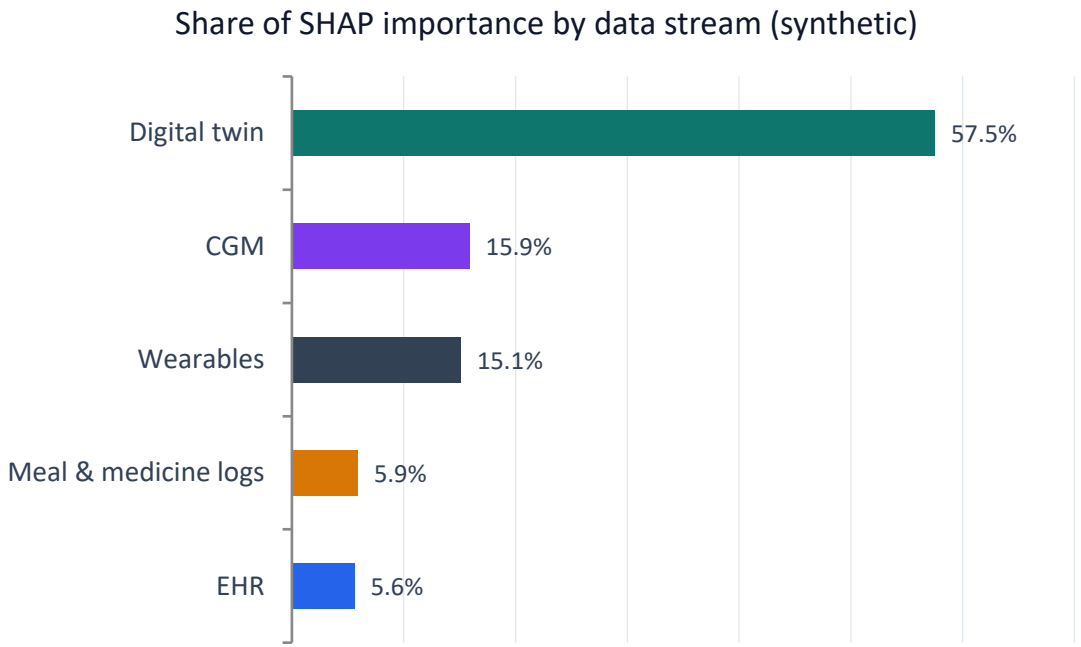
83.9%

Clarke zone A at 60 min



What the real data tells us (honestly): the same pipeline transfers to real, noisy, hospital CGM and beats rule-based and no-change baselines by a wide margin. This dataset has no wearables and free-text diet logs, so EHR and twin features add little beyond CGM here - the fusion gains we see on synthetic data still need prospective validation with Indian patients wearing CGM + smartwatch.

Explainability: every alert comes with its reasons



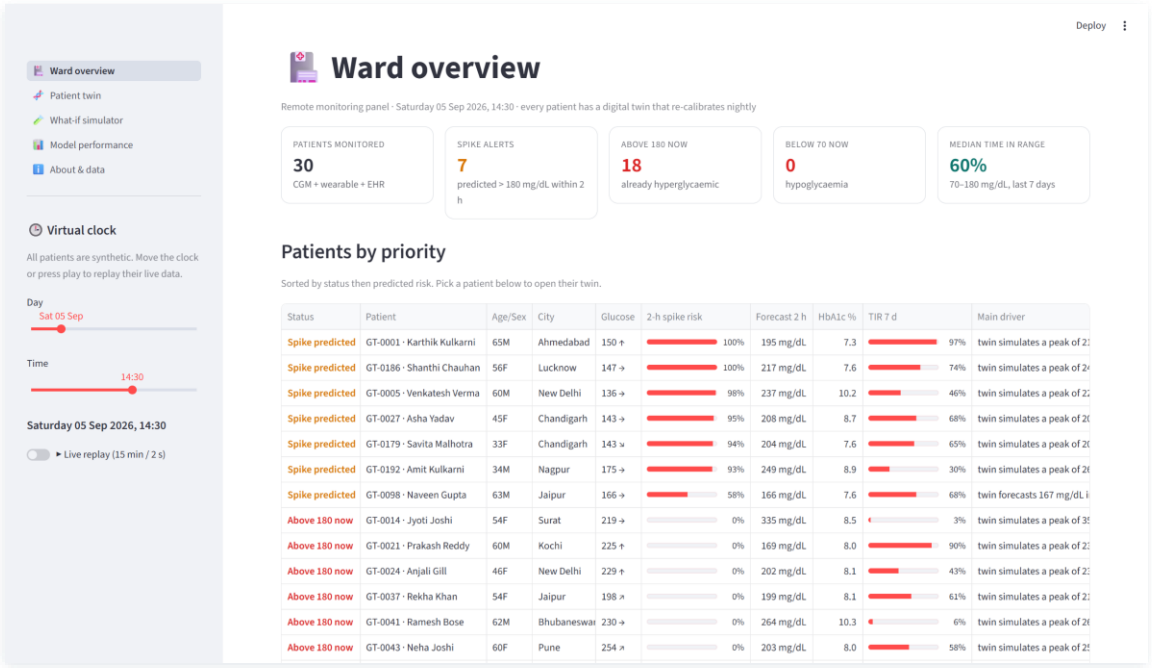
The twin's simulation carries most of the signal, followed by wearables and CGM history - the fusion is what the model actually uses.

Example alert · GT-0001 · Sat 05 Sep, 13:55

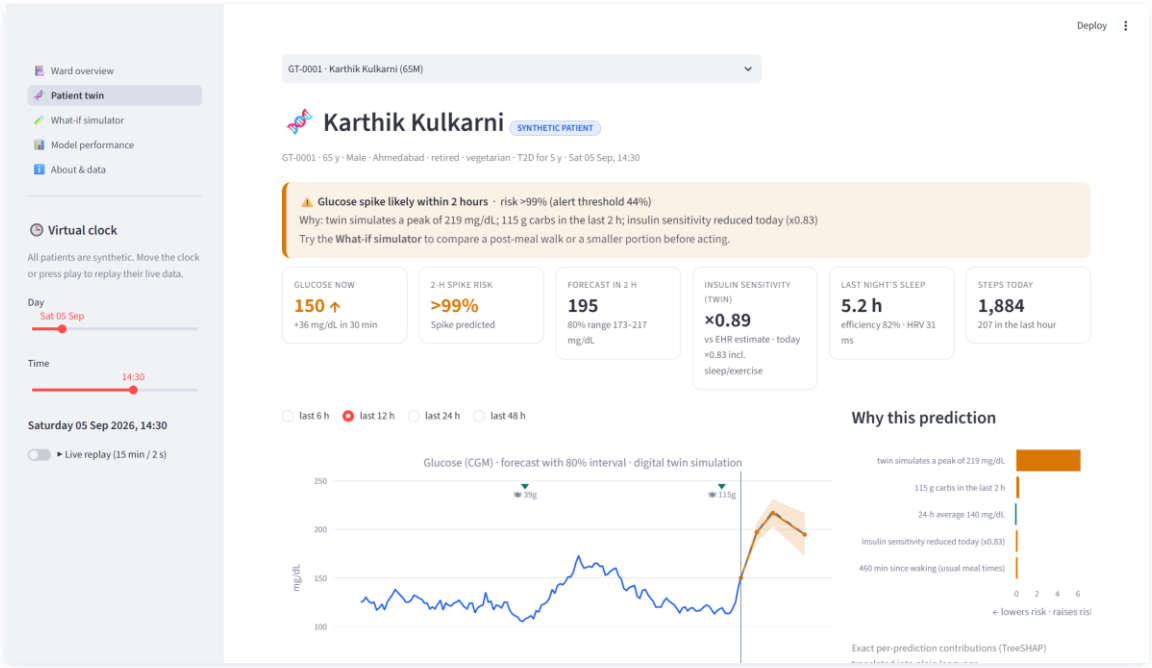
Glucose 119 mg/dL · spike risk 99% · forecast in 2 h 212 mg/dL

- twin simulates a peak of 221 mg/dL (raises risk)
- 115 g carbs in the last 2 h (raises risk)
- twin learned low insulin sensitivity (x0.89) (raises risk)
- 425 min since waking (usual meal times) (raises risk)

The doctor's view: a ward of digital twins



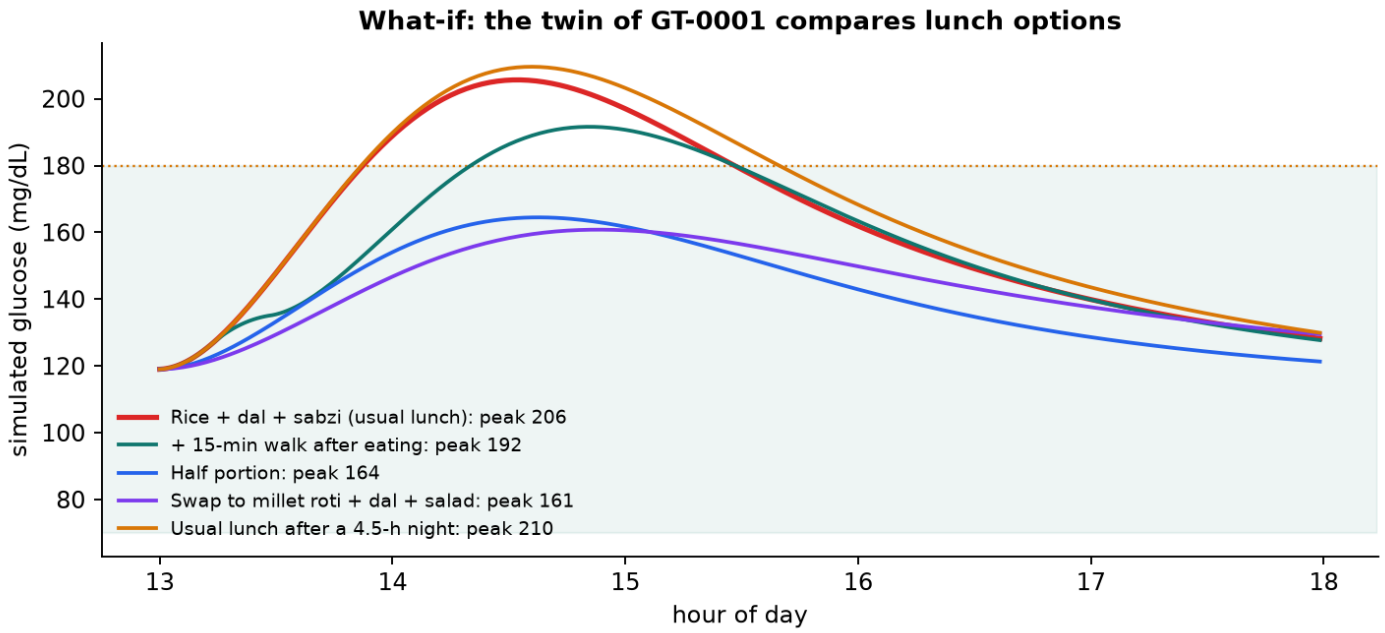
Ward overview: every patient ranked by 2-h spike risk



Patient twin: CGM, forecast band, alert reasons, wearables, EHR

Also: what-if simulator, model-performance page, live replay mode and a FastAPI REST API (/patients, /state, /twin, /what-if).

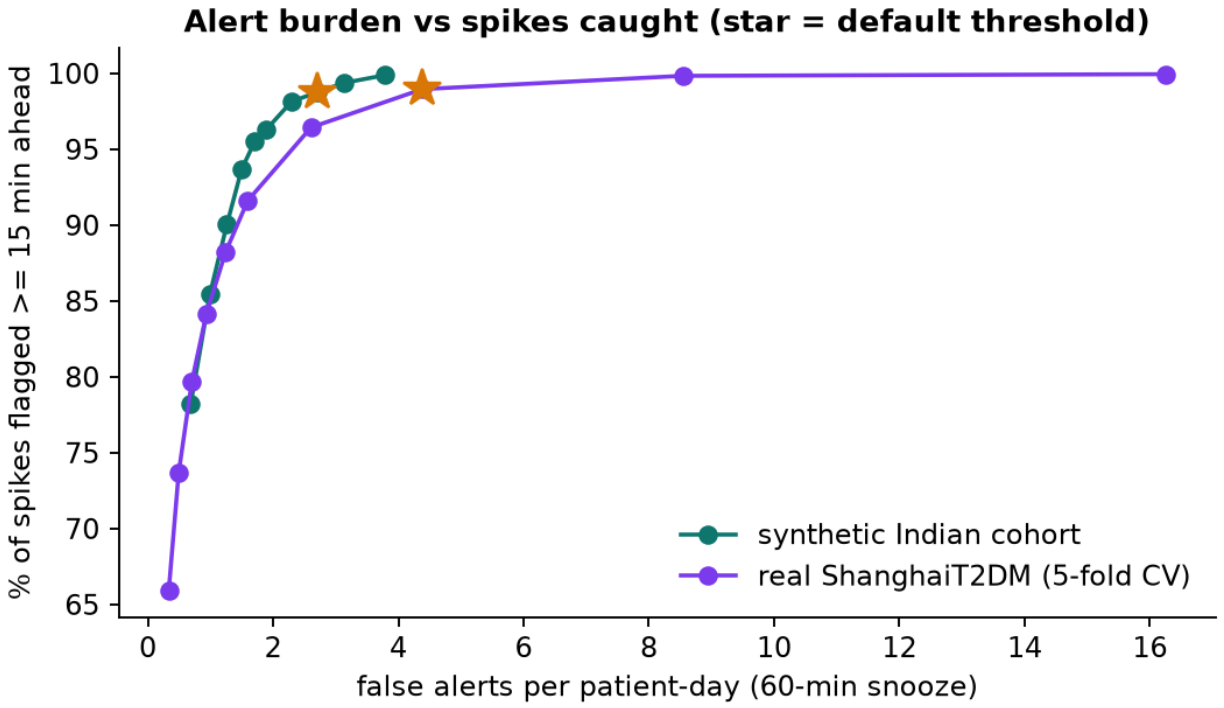
What-if: test the plan on the twin before the patient eats



Scenario	Peak	vs usual
Rice + dal + sabzi (usual lunch)	206 mg/dL	-
+ 15-min walk after eating	192 mg/dL	-14
Half portion	164 mg/dL	-42
Swap to millet roti + dal + salad	161 mg/dL	-45
Usual lunch after a 4.5-h night	210 mg/dL	+4

Simulated with the patient's own calibrated twin - so the advice is personal: a walk helps some patients more than a swap, and vice versa.

Choosing the alert setting: sensitive vs quiet



Sensitive (default)
99% of spikes flagged
median 110 min warning
2.7 false alerts / day

Quiet
85% of spikes flagged
median 70 min warning
1.0 false alerts / day

Synthetic cohort; repeat notifications snoozed for 60 min. Care teams can pick the operating point that fits their capacity.

Privacy, safety and honest limitations

Privacy by design

- Only synthetic or published anonymised data (sandbox rules)
- No identifiers in features; FHIR bundles tagged synthetic
- DPDP Act 2023 principles: purpose limitation, data minimisation

Clinical safety

- Decision support for a care team, never automated dosing
- Calibrated 80% intervals shown with every forecast
- Alerts carry reasons so clinicians can disagree

Limitations

- Wearable-glucose effects in the synthetic cohort come from literature effect sizes
- Real-data track lacks wearables; carbs estimated from free text
- Needs prospective validation in Indian clinics before any clinical use

0 identifiable records used. 200 synthetic patients + 100 published anonymised patients (CC BY 4.0). GlucoTwin is a research prototype, not a medical device.

From prototype to pilot



Pilot

3-month IEC-approved pilot with CGM + smartwatch users at an Indian diabetes clinic

Integrate

ABDM / FHIR connectors, Libre & Health Connect ingestion, WhatsApp alerts for care teams

Scale

Federated learning across hospitals so data never leaves the site

Extend

Add hypertension and cardiovascular twins on the same engine and dashboard

Impact goal: turn the CGM data India's patients already generate into early, explained, personal action - fewer hours above range, without more work for clinicians.

Thank you

GlucoTwin - Team Gym Bro, IIIT Delhi

Code, data card, model card and video: see the GitHub repository README
Run it: python run_all.py · streamlit run dashboard/app.py
MIT licence · ShanghaiT2DM data CC BY 4.0 (Zhao et al., Sci Data 2023)